

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1082

West -Station 138 kV
July 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1082. The Interconnection Customer proposes a 151.4 MW battery energy storage facility interconnecting at the West -Station 138 kV substation in the Maryland area. The request was received on October 18, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 319 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 28 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,026,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

ST BONIFACIUS – W -STATION 138 KV: 135.9% loading (1,496 MVA) following the outage of the SOUTHTOWN – VIENNA OPERATIONS 345 KV; DFAX = 72.0%.

For the loss of SOUTHTOWN SUBSTATION – VIENNA OPERATIONS 345 KV, the ARROYO TAP – MESQUITE 138 KV loads to 133.2% of its 1,106 MVA rating. Project DFAX: 27.9%.

The ARROYO TAP – REIDLAND 138 KV is overloaded to 132.5% of its 1,126 MVA emergency rating for the loss of the VIENNA OPERATIONS – SOUTHTOWN 345 KV. The project contributes 27.9% DFAX (42.2 MW) to this overload.

OCEAN CITY – REIDLAND 138 KV: 132.5% loading (1,126 MVA) following the outage of the SOUTHTOWN – VIENNA OPERATIONS 345 KV; DFAX = 27.9%.

For the loss of SOUTHTOWN – VIENNA OPERATIONS 345 KV L, the OCEANCITY – CANDELARIA ARENAS 1718 138 KV Line loads to 131.9% of its 1,141 MVA rating. Project DFAX: 27.9%.

For the loss of VIENNA OPERATIONS – EASTON 2 345 KV, the ANGOLA – ROBINSONVILLE 138 KV L loads to 131.3% of its 180 MVA rating. Project DFAX: 6.9%.

The CADY 345/138 kV Auto is overloaded to 131.0% of its 823 MVA emergency rating for the loss of the DODD ROAD – CADY 345 KV line. The project contributes 22.9% DFAX (34.7 MW) to this overload.

CADY – KEY LARGO 138 KV line: 130.3% loading (214 MVA) following the outage of the CADY – DODD ROAD 345 KV; DFAX = 5.5%.

The OAK HILL – LOUISA SUBSTATION 138 KV line is overloaded to 129.8% of its 324 MVA emergency rating for the loss of the DODD RD – CADY 345 KV. The project contributes 9.6% DFAX (14.5 MW) to this overload.

The CADY – LOUISA 138 KV is overloaded to 128.1% of its 338 MVA emergency rating for the loss of the CADY – DODD ROAD 345 KV. The project contributes 9.6% DFAX (14.5 MW) to this overload.

For the loss of EASTON 2 – VIENNA OPERATIONS 345 KV, the NEWMARK – ESS LEWES SUBSTATION 138 KV loads to 126.1% of its 515 MVA rating. Project DFAX: 21.0%.

The LOST HILLS – OAK HILL 138 KV is overloaded to 125.6% of its 466 MVA emergency rating for the loss of the CADY – DODD RD 345 KV. The project contributes 14.5% DFAX (21.9 MW) to this overload.

WYE MILLS – HOLSTON DAM 138 KV: 124.7% loading (231 MVA) following the outage of the EASTON 2 – CADY 345 KV line; DFAX = 5.8%.

SANDBURG – JARRATT 138 KV: 123.4% loading (715 MVA) following the outage of the VIENNA OPERATIONS – EASTON 2 345 KV; DFAX = 27.4%.

The NORTH LAKE SUBSTATION – KIWI 138 KV line is overloaded to 122.8% of its 688 MVA emergency rating for the loss of the VIENNA OPERATIONS – EASTON 2 345 KV L. The project contributes 23.7% DFAX (35.9 MW) to this overload.

For the loss of VIENNA OPERATIONS – EASTON 2 345 KV, the SANDBURG – SHERWOOD 138 KV loads to 122.8% of its 727 MVA rating. Project DFAX: 27.4%.

SHERWOOD – HITCHLAND 138 KV: 122.4% loading (734 MVA) following the outage of the EASTON 2 – VIENNA OPERATIONS 345 KV; DFAX = 27.4%.

HILLSBORO – PARK AVE 138 KV Line: 122.0% loading (760 MVA) following the outage of the CADY – EASTON 2 345 KV; DFAX = 25.1%.

For the loss of VIENNA OPERATIONS – EASTON 2 345 KV Line, the HITCHLAND SUBSTATION – N LAKE 138 KV loads to 121.1% of its 764 MVA rating. Project DFAX: 27.4%.

The CHEMOLITE – RANDALL COUNTY 138 KV is overloaded to 120.2% of its 399 MVA emergency rating for the loss of the VIENNA OPERATIONS – EASTON 2 SUBSTATION 345 KV. The project contributes 16.7% DFAX (25.2 MW) to this overload.

The PARK AVE – HAROR 138 KV Line is overloaded to 119.7% of its 410 MVA emergency rating for the loss of the EASTON 2 – CADY 345 KV. The project contributes 9.5% DFAX (14.4 MW) to this overload.

CHEMOLITE – LAMMERS 138 KV: 119.1% loading (413 MVA) following the outage of the VIENNA OPERATIONS – EASTON 2 345 KV; DFAX = 16.7%.

For the loss of CADY – EASTON 2 345 KV, the HARBOR – HOLSTON DAM 138 KV line loads to 118.1% of its 432 MVA rating. Project DFAX: 9.5%.

The MACON – GRASONVILLE 138 KV is overloaded to 117.3% of its 433 MVA emergency rating for the loss of the CADY – EASTON 2 345 KV. The project contributes 11.3% DFAX (17.1 MW) to this overload.

CARANCAHUA – HOLSTON DAM 138 KV: 117.2% loading (235 MVA) following the outage of the CADY – EASTON 2 345 KV; DFAX = 5.5%.

W -STATION SUBSTATION – SHINGLE CRK 138 KV line: 102.4% loading (1,370 MVA) following the outage of the SOUTHTOWN – VIENNA OPERATIONS 345 KV; DFAX = 28.0%.

SHINGLE CRK – SOUTHTOWN 138 KV: 102.0% loading (1,403 MVA) following the outage of the SOUTHTOWN – VIENNA OPERATIONS 345 KV; DFAX = 28.0%.

The LANGLOIS – VEDDER 138 KV is overloaded to 101.4% of its 750 MVA emergency rating for the loss of the COMMONWEALTH CHESAPEAKE – SOUTHTOWN 345 KV. The project contributes 10.3% DFAX (15.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ST BONIFACIUS – W -STATION 138 KV	Rebuild line with higher-capacity conductor	\$5,283,000	5% (\$252,000)
ARROYO TAP – MESQUITE 138 KV	Rebuild line with higher-capacity conductor	\$3,002,000	2% (\$76,000)
ARROYO TAP – REIDLAND 138 KV	Rebuild line with higher-capacity conductor	\$1,335,000	2% (\$34,000)
OCEAN CITY – REIDLAND SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$5,128,000	2% (\$130,000)
OCEANCITY – CANDELARIA ARENAS 1718 138 KV Line	Rebuild line with higher-capacity conductor	\$3,844,000	2% (\$97,000)
ROBNSONVILLE – ANGOLA 138 KV L	Rebuild line with higher-capacity conductor	\$5,142,000	3% (\$170,000)
CADY 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$8,898,000	3% (\$257,000)
CADY – KEY LARGO 138 KV	Rebuild line with higher-capacity conductor	\$9,623,000	3% (\$263,000)
OAK HILL – LOUISA SUBSTATION 138 KV line	Rebuild line with higher-capacity conductor	\$2,950,000	3% (\$88,000)
CADY – LOUISA 138 KV	Rebuild line with higher-capacity conductor	\$3,934,000	3% (\$117,000)
ESS LEWES SUBSTATION – NWMARK 138 KV line	Rebuild line with higher-capacity conductor	\$14,717,000	3% (\$477,000)
LOST HILLS – OAK HILL 138 KV	Rebuild line with higher-capacity conductor	\$14,757,000	3% (\$448,000)
WYE MILLS – HOLSTON DAM 138 KV line	Rebuild line with higher-capacity conductor	\$6,673,000	3% (\$206,000)
JARATT – SANDBURG 138 KV line	Rebuild line with higher-capacity conductor	\$6,993,000	4% (\$270,000)
NORTH LAKE SUBSTATION – KIWI 138 KV line	Rebuild line with higher-capacity conductor	\$6,679,000	4% (\$247,000)
SANDBURG – SHERWOD 138 KV	Rebuild line with higher-capacity conductor	\$2,874,000	4% (\$111,000)
SHERWOOD – HITCHLAND 138 KV	Rebuild line with higher-capacity conductor	\$9,777,000	4% (\$377,000)
PARK AVE – HILLSBORO 138 KV L	Rebuild line with higher-capacity conductor	\$5,314,000	3% (\$183,000)
HITCHLAND SUBSTATION – N LAKE 138 KV	Rebuild line with higher-capacity conductor	\$6,463,000	4% (\$249,000)
CHEMOLITE – RANDALL COUNTY 138 KV	Rebuild line with higher-capacity conductor	\$7,924,000	4% (\$324,000)
HARBOR – PARK AVE 138 KV L	Rebuild line with higher-capacity conductor	\$11,300,000	3% (\$298,000)
CHEMOLITE – LAMMERS 138 KV	Rebuild line with higher-capacity conductor	\$5,014,000	4% (\$205,000)
HARBOR – HOLSTON DAM 138 KV	Rebuild line with higher-capacity conductor	\$15,367,000	3% (\$405,000)
MACON – GRASONVILLE 138 KV	Rebuild line with higher-capacity conductor	\$2,647,000	3% (\$80,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HOLSTON DAM – CARANCAHUA 138 KV	Rebuild line with higher-capacity conductor	\$13,036,000	3% (\$403,000)
W -STATION – SHINGLE CRK 138 KV	Reconductor line	\$4,442,000	2% (\$96,000)
SHINGLE CRK – SOUTHTOWN 138 KV Line	Reconductor line	\$1,348,000	2% (\$29,000)
LANGLOIS – VEDDER 138 KV	Reconductor line	\$7,984,000	2% (\$134,000)
	Total allocated to project		\$6,026,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.